

A joint model for the surprise examination and temporal knower sentences

On the concluding conjecture of Aldini, Alexander and Graziani,
Knowledge-of-own-factivity, the definition of surprise, and a solution to the Surprise Examination paradox

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Status. A proposed resolution, by a complete hand argument. Independent review and external priority checking remain. No proof-assistant check has been run.

Source. A. Aldini, S. A. Alexander and P. Graziani, *Knowledge-of-own-factivity, the definition of surprise, and a solution to the Surprise Examination paradox*, CIFMA 2022, Springer LNCS, 2023. [doi:10.1007/978-3-031-26236-4_30](https://doi.org/10.1007/978-3-031-26236-4_30). The text examined is the 17-page conference pre-proceedings version ([PDF](#)).

What is shown. The paper conjectures that its temporal theory of knowledge can be extended by self-referential sentences L_i (“this sentence is known at time i to be false”). It further asks that every later time knows the falsehood of every earlier L_i , and that the whole remain consistent. A model is constructed. It preserves:

- factivity, retention and deduction;
- knowledge of logical validities;
- delayed knowledge of factivity;
- stratified closure.

It adds the non-atomic sentences L_i with their exact semantics, and satisfies the stronger demand. This is not a Kripke/S4 model.

Contribution. The paper’s §5 construction supplies the main method. What is new here is:

- the extension to its proposed self-referential language;
- an explicit closure argument;
- the observation that the additional delayed-knowledge condition follows automatically (§2).

Check script: `temporal_knower_controls.py` in this folder, which writes `temporal_knower_controls.json`.

1. Language and the semantic point that matters

Fix an examination window of $n \geq 3$ days. First use active temporal indices $I = \{1, \dots, n\}$. As in the source, the **language** still has D_i and K_i for every positive integer i , together with the usual Boolean connectives and the new non-atomic L_i for every positive i . Only the outer temporal indices of the theory’s schemas are restricted to I ; their formula parameters may mention indices beyond the examination window.

Following the paper, a model freely assigns truth values to basic formulas D_i and $K_i\varphi$. It evaluates Boolean connectives classically, and the added clause is

$$M \models L_i \iff M \models K_i \neg L_i.$$

This is well-defined: $K_i \neg L_i$ is a **basic formula** with an assigned truth value. Evaluating it does not recursively evaluate its argument. In particular, L_i is neither an independently valued propositional atom nor a string to be expanded infinitely inside every modality.

For precision, translate each $K_i \varphi$ to a distinct propositional atom $a_{i,\varphi}$, each D_i to d_i , and L_i to $a_{i,\neg L_i}$; translate Boolean connectives homomorphically. This translates satisfaction exactly to ordinary propositional satisfaction. Consequently semantic consequence is compact, has the usual finite-premise deduction property, and includes the validity $L_i \leftrightarrow K_i \neg L_i$. These are validities of the expanded semantics, as required by the source's definition of tautology. All schemas below range over the **expanded** language.

Write $\text{Cn}(T)$ for semantic consequence in this language, and $F_i = \{K_i \varphi \rightarrow \varphi : \varphi \text{ any formula}\}$ for the factivity schema at time i .

2. The stronger delayed-knowledge condition is automatic once consistency holds

Put $q_i = (K_i \neg L_i \rightarrow \neg L_i)$. By the semantic clause,

$$q_i \rightarrow \neg L_i$$

is valid: if L_i were true, both $K_i \neg L_i$ and L_i would be true, making q_i false. Any factive model satisfies q_i , hence $\neg L_i$.

If $j > i$, delayed knowledge of factivity includes $K_j q_i$. Knowledge of validities gives $K_j(q_i \rightarrow \neg L_i)$, and closure under modus ponens gives

$$K_j \neg L_i.$$

Thus the additional conjectural requirement follows from the proposed semantics and existing schemas. The main problem is producing a joint model of those schemas; no extra axiom asserting $K_j \neg L_i$ is needed.

3. Spell out the stratified theories

Let S contain the following source axioms, with all indicated formulas in the expanded language:

1. $\bigvee_{i=1}^n D_i$, and $\neg(D_i \wedge D_j)$ for $1 \leq i < j \leq n$.
2. The paper's broadened surprise assertion

$$\bigvee_{i=1}^n (D_i \wedge \neg K_i D_i) \vee \bigvee_{1 \leq i < j \leq n} (D_i \wedge K_i D_j).$$

3. $\neg D_i \rightarrow K_{i+1} \neg D_i$ for $i < n$.
4. $K_i \varphi$ for each valid φ and $i \in I$.
5. $K_i(\varphi \rightarrow \psi) \rightarrow (K_i \varphi \rightarrow K_i \psi)$.
6. $K_i \varphi \rightarrow K_j \varphi$ for $i < j$.
7. $K_j(K_i \varphi \rightarrow \varphi)$ for $i < j$.

For $J \subseteq I$, let $\text{Cl}_J(T)$ be the **least** theory containing T , closed under semantic consequence and under $\varphi \mapsto K_j \varphi$ for each $j \in J$. Such a theory exists by intersection of all closed supersets. Compactness also permits construction by iterating the closure operations through finite stages and taking their union. This is a definition of sets of formulas, not a claim that their membership or semantic consequence is decidable.

Define

$$B_0 = \text{Cl}_I(S), \quad B_i = \text{Cl}_{\{i\}} \left(B_{i-1} \cup \bigcup_{r < i} F_r \right) \quad (1 \leq i \leq n).$$

In particular $B_1 = B_0$. These are the deductively closed presentations of the paper's $(V_n)_0$ and $(V_n)_0^i$ in the expanded language. Closing under consequence changes no models. To check the correspondence, the new part of stage i is precisely past factivity and knowledge, at time i , of that stage's consequences. The lower-time closure clauses are already included in B_{i-1} . Thus $B_i \subseteq B_j$ for $i < j$, and whenever $B_i \models \varphi$, $K_i \varphi \in B_i$.

The desired full theory is

$$V_n^L = \bigcup_{i=1}^n B_i \cup \bigcup_{i=1}^n F_i.$$

The final factivity schemas are **not** placed inside every earlier knowledge closure. Doing so would change the theory and restore the knower obstruction.

4. One auxiliary model proves the needed ignorance

Take the actual exam to be on day 1. To ensure it is not known beforehand, we first show $B_0 \not\models D_1$.

Define N by making D_2 true and every other D_i false, and assigning **every** basic $K_i \varphi$ truth value true. The added semantic clause therefore makes every L_i true. It would be an error to declare these L_i false while keeping all knowledge formulas true.

N satisfies S . The examination is unique. The second surprise disjunction has the witness $D_2 \wedge K_2 D_3$, available because $n \geq 3$. The observation, deduction, retention, knowledge-of-validity and delayed-factivity clauses hold because all their modal conclusions are true. Moreover every formula added by a knowledge-closure rule is true in N , and semantic consequence preserves truth in N . Hence $N \models B_0$, but $N \not\models D_1$.

The auxiliary model is deliberately nonfactive. It witnesses non-entailment by a theory that has not yet assumed current/future factivity. It is not the final model and does not weaken the final factivity requirement.

5. The actual model

Let

$$H_i = \{\neg D_j : 1 \leq j < i, j \neq 1\}, \quad W_i = B_i \cup H_i.$$

The W_i are nested. Assign the basic truth values of M by

$$M \models D_j \iff j = 1,$$

$$M \models K_i \varphi \iff W_i \models \varphi, \quad i \in I.$$

For positive $i \notin I$, assign every basic $K_i \varphi$ false. This completes the valuation on the full source language. The finite-window theory imposes no factivity or closure schemas at those outer indices, though formulas containing them remain available inside all active-index schemas.

The truth values of the L_i are supplied by their semantic clause, not chosen separately. All theories on the right were defined before M ; this is a mathematical model definition, not an effective evaluation algorithm.

First verify $M \models S$:

- The exam is unique. Since $W_1 = B_1 = B_0 \not\models D_1$, the surprise witness is $D_1 \wedge \neg K_1 D_1$.
- If $M \models \neg D_j$, then $j \neq 1$ and $\neg D_j \in H_{j+1}$, giving the observation clause.
- Valid formulas are consequences of every W_i . Semantic consequence is closed under modus ponens, giving modal deduction.
- $W_i \subseteq W_j$ gives retention.
- If $r < i$, then $F_r \subseteq B_i \subseteq W_i$. Thus $M \models K_i(K_r \varphi \rightarrow \varphi)$, giving delayed factivity.

Every application of a knowledge-closure rule while forming B_0 introduces a true formula in M : for all i , W_i contains B_0 , so $B_0 \models \varphi$ implies $M \models K_i \varphi$. Together with $M \models S$, closure induction gives $M \models B_0$.

Now prove $M \models B_i$ and $M \models W_i$ by induction on i . All H_i are true under the chosen actual exam date. Suppose the earlier W_r are true in M . For each $r < i$, if $M \models K_r \varphi$, the definition says $W_r \models \varphi$; because $M \models W_r$, it follows that $M \models \varphi$. Thus $M \models F_r$ for every $r < i$.

The seeds $B_{i-1} \cup \bigcup_{r < i} F_r$ of B_i therefore hold in M . Any formula $K_i \varphi$ introduced by its knowledge-closure rule is also true: $B_i \models \varphi$ implies $W_i \models \varphi$ by inclusion. Closure induction gives $M \models B_i$, and the true histories give $M \models W_i$. This argument does not assume current factivity to prove current soundness: only earlier W_r are used in the factivity step.

Finally, for every i and φ , if $M \models K_i \varphi$, then $W_i \models \varphi$ and $M \models W_i$, hence $M \models \varphi$. Therefore M is factive at every time and satisfies V_n^L .

By §2 it also satisfies, for every $i < j \leq n$,

$$\neg L_i, \quad \neg K_i \neg L_i, \quad K_j \neg L_i.$$

The middle statement follows from the defining equivalence and $\neg L_i$. The additional later-knowledge conjecture is thus satisfied in the same model.

6. All positive times, and the exact boundary of the result

If “whenever $i < j$ ” is intended to quantify over **all positive times**, the same construction works. Keep the examination window $1, \dots, n$ with $n \geq 3$, but take $I = \mathbb{N}_{>0}$. Include deduction, retention, knowledge of validities and delayed factivity for all temporal indices, and observation $\neg D_i \rightarrow K_{i+1} \neg D_i$ for every positive i . Define B_i, H_i, W_i for each finite i and take the union of all B_i and all F_i as the final theory. Use the same N and M , with all D_j except the chosen examination day false, including days beyond the window. Every argument above is indexed by a finite i and remains valid. This gives $K_j \neg L_i$ for all positive $i < j$, with an examination still held within the original finite window. This is a stated stronger temporal extension, not a silent change to the paper’s finite-window theory.

For $n = 1, 2$ the source already proves its original theory inconsistent. Those derivations use only the original language and remain available here; the extension does not repair them. The threshold of three days is inherited from the source, not a new result claimed by this note.

Adding $K_i q_i$, knowledge at time i of its own particular factivity instance, would be inconsistent with these requirements. The validity $q_i \rightarrow \neg L_i$, known deduction and $K_i q_i$ would give $K_i \neg L_i$, therefore L_i , contradicting factivity. The proof identifies the missing knowledge precisely; it does not abandon actual factivity or retention.

Verification and source limits

The script checks:

- the special semantic translation;
- the short derivation from factivity to falsehood;
- the obstruction from same-time knowledge of factivity;
- finite instances in the all-knowing auxiliary valuation.

It also includes a negative control showing why setting the new sentences false independently breaks the auxiliary model. These finite tests do not check the infinite theory closures; §§3–5 supply that hand argument.

The source’s Definition 4 was inspected visually on printed p. 10. Its §5 proof and concluding conjecture on p. 16 were read. A targeted public search on 21 September 2026 did not locate a later explicit solution of this joint-model conjecture. No comprehensive citation search was available, so priority is unresolved.

References

- A. Aldini, S. A. Alexander, P. Graziani, as above (Definition 4, §5, and the concluding conjecture).